Anmol Sikka

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Research Interests: Planetary Regolith Mechanics & Terramechanics, Discrete Element Modeling (DEM), Spacecraft Mission Design
Teaching Interests: Aerospace Design, Space Systems Engineering, Astrodynamics, Computational Methods for Aerospace Engineering

Education

2026 **PhD, Aerospace Engineering**, *University of Maryland, College Park*, Advisor: *Prof. Christine Hartzell*, Thesis: *Role of Magnetic Forces on Regolith Behavior on Metallic Asteroids with a Residual Magnetic Field*.
Major: Flight Dynamics and Control; Minor: Space Systems

2020 **MS, Aerospace Engineering**, *Indian Institute of Technology (IIT) Bombay*, Advisor: *Prof. Ravi Banavar*, Thesis: *Discrete mechanics and optimal control techniques applied to a reentry launch vehicle*.

2020 **BS, Aerospace Engineering**, *Indian Institute of Technology (IIT) Bombay*.

Research Experience

2026 - present **Postdoctoral Associate** *Advisor: Prof. Christine Hartzell, UMD*

- Conducting DEM modeling of magnetic granular avalanches on asteroid (16) Psyche to investigate density-driven segregation and magnetic accumulation in crater basins
- Quantifying plasma wave (soliton) detection rates for multi-spacecraft formation architectures to detect orbital debris in LEO
- Mentoring incoming graduate researchers on orbital debris detection workflows

2025 - 2026 **Graduate Research Assistant** *Advisor: Prof. Christine Hartzell, UMD*

- Modelled CubeSat orbital trajectories to optimize sensor placement using Monte Carlo simulations of the LEO debris environment to maximize detection probability for NASA's NIAC Phase II Project

2021 - 2026 **Doctoral Thesis** *Advisor: Prof. Christine Hartzell, UMD*

- *Geophysical Characterization*: Conducted multi-scenario avalanching simulations to map the conditions on asteroid 16-Psyche where the magnetic forces dictate the surface morphology *(Sikka & Hartzell 2026)*
- *Computational Framework*: Developed the first experimentally validated open-source DEM framework in LIGGGHTS to simulate cohesive and magnetic regolith dynamics in low-gravity environments, establishing critical time-step formulations for multi-body contact stability *(Sikka & Hartzell 2026)*
- *Analytical Modeling*: Formulated an empirical force model for inter-particle interactions between paramagnetic grains in uniform magnetic fields, tailored for M-type asteroids *(Sikka et al. 2023)*

2019 - 2020 **Master's Thesis** *Advisor: Prof. Ravi Banavar, IIT Bombay*

- Formulated glide-type launch-vehicle re-entry as an optimal control to minimize the total thermal flux for Indian Space Research Organization (ISRO)'s Reusable Launch Vehicle (RLV-TD) Project
- Compared direct optimization (CasADi in Python) and indirect optimization (Discrete-time Maximum Principle) methods while minimizing the total thermal flux on the vehicle

2018 **Bachelor's Thesis** *Advisor: Prof. Ashok Joshi, IIT Bombay*

- Implemented and optimised the Differential Evolution (DE), a genetic algorithm to solve Lambert's problem for a perturbed orbit with an accuracy of determined orbital elements within 10 centimeters

Summer 2017 **Summer Research Intern** *Advisor: Prof. R. V. Ramanan, IIST, Thiruvananthapuram*

- Analysed finite-burn maneuver parameters like starting position, burn time and steering techniques to optimize the achieved orbit under minimum fuel consumption

Relevant Expertise

Computational Physics & Granular Mechanics: Discrete Element Method (DEM), LIGGGHTS, Granular Contact Mechanics, Multiphysics Coupling (Magnetic/Cohesive Forces)

High-Performance Computing & Code: C++, Python (NumPy, SciPy), MPI Parallel Computing, Linux/Bash, Git, High-Performance Clusters (HPC)

Teaching Topics Engineering Mechanics, Dynamics, Design under Constraints, Systems Engineering

Hands-on Engineering Vibration Testing, Thermal-Vacuum Test Plan, Prototype Integration & Validation

Journal Articles

2026 **A.Sikka**, C.M.Hartzell. *Investigating the Role of Magnetic Cohesion in Avalanching on 16-Psyche: A Discrete Element Modeling Approach.*, Planet. Sci. J. 7 224, DOI: 10.3847/PSJ/aea4b0.

2026 **A.Sikka**, C.M.Hartzell. *An Experimentally Validated Magnetic Force Model for Discrete Element Modeling of Paramagnetic Granular Media.*, Granular Matter, 28, 53, DOI: 10.1007/s10035-026-01643-x

2023 **A.Sikka**, I.DesJardin, T.Leps, and C.Hartzell, *Development of an Empirical Model of the Force between Paramagnetic Particles in Uniform Magnetic Field on M-type Asteroids*, Planet. Sci. J. 4, 129, DOI: 10.3847/PSJ/ace323.

White Papers

2024 M.Hirabayashi et al. (incl. **A.Sikka**) *Near-Earth Objects Workshops to Assess Reconnaissance for Planetary Defense: Working Groups Final Report.* LINK: ntrs.nasa.gov/citations/20250005751

Conference Proceedings

Oral Presentations

2024 **A.Sikka** and C.Hartzell. *DEM Simulations of avalanching of metallic regolith under the influence of magnetic forces*. At 56th Annual Meeting of the Division for Planetary Sciences (DPS), Oct. 2024

2024 **A.Sikka** and C.Hartzell. *Extending LIGGGHTS to Perform Discrete Element Method Simulations of Mass Wasting Events in Uniform Magnetic Field on Psyche (16)-Like Metallic Bodies. At 55th LPSC, Mar. 2024*

2022 **A.Sikka**, I.DesJardin and C.Hartzell. *Characterizing Magnetic Force Between Paramagnetic Particles of Magnetic Asteroids*. At 53rd Lunar Planetary Science Conference (LPSC), Mar. 2022

Poster Presentations

2024 **A.Sikka** and C.Hartzell. *Magnetic Force Model to simulate Avalanching on Metallic Asteroids like 16-Psyche*. At Granular Matter Gordon Research Seminar (GRS) and Conference (GRC), Jun. 2024

2023 **A.Sikka** and C.Hartzell. *Empirical model of the force between paramagnetic particles in uniform magnetic field on m-type asteroids*. At Asteroids, Comets and Meteors (ACM) Conference, Jun. 2023

Second Author (*equal contribution)

2021 A.R.Ranade*, **A.Sikka*** et al. *Survey and Analysis of Payloads for Missions on PSLV's Orbital Platform*. In Scitech Forum, American Institute of Aeronautics and Astronautics (AIAA), 2021

2020 Y.Sanghvi*, **A.Sikka*** and A.R. Ranade. *Quality Assurance Practices for Student Satellite Teams*. In Advances in Small Satellite Technologies, Springer, Singapore, 2020, pp. 439–450.

Other Author

2024 I.Mishra et al. (incl. **A.Sikka**) *A Concept for a New Frontiers Class Ceres Lander*. At the annual American Geophysical Union (AGU) meeting, Dec. 2024

2023 M.Buys et al. (incl. **A.Sikka**) *SIZLE: Smallsat to Image Zodiacal Light Above Ecliptic*. in ASCEND, American Institute of Aeronautics and Astronautics (AIAA), Oct. 2023

2020 K.Jagdale et al. (incl. **A.Sikka**) *Sanket - Technology Demonstration of Antenna Deployment System on PSLV Stage 4 Orbital Platform*.
At Small Satellite Technology & Applications National Conference, 2020

2019 A.R.Ranade, et al. (incl. **A.Sikka**) *Learnings from Pratham, First Student Satellite of IIT Bombay*.
At International Conference on Small Satellites (ICSS), Feb. 2019

Awards and Fellowships

Fall 2024 **Dean's Fellowship** — *Aerospace Engineering, University of Maryland for 2024-25 Academic Year*

Spring 2024 **Future Faculty Fellow** — *A. James Clark School of Engineering, University of Maryland*

2020 **Institute Technical Citation** — *Technical Council of IIT Bombay*

2019 **2nd place at CubeSat Design Challenge** — *International Conference on Small Satellites (ICSS)*

2019 **Institute Technical Color** — *Technical Council of IIT Bombay*

2015 **Kishore Vaigyanik Protsahan Yojna (KVPY) Fellowship** — *Department of Science & Technology, India*

2013 **National Talent Search Examination (NTSE) Fellowship** — *NCERT, India*

2011 **2nd place at University of Toronto Space Design Contest (UTSDC)** — *Held in Toronto, Canada*

Engineering Experience

Summer 2024 **CORA: Ceres Ocean Reconnaissance and Analysis** *JPL's Planetary Science Summer School (PSSS)*

- Collaborated on a 17-person concurrent engineering team to develop a New Frontiers-class lander/hopper mission concept targeting volatile-rich surface deposits in Ceres' Occator Crater
- Interfaced with science and engineering subsystems to formulate the Science Traceability Matrix (STM), defining surface payload requirements for in-situ regolith characterization and volatile detection
- Served as Cost Lead alongside JPL's Team X, pricing and balancing the full mission lifecycle using JPL Institutional Cost Models (ICM) across NASA Work Breakdown Structures (WBS) within designated cost caps

Feb. 2023 **Lucy Occultations Campaign** *Southwest Research Institute (SWRI)*

- Trained to set up & operate a CPC1100 telescope at remote locations (Deployed in KS for the campaign)
- Confirmed detection of Asteroid Polymele, Lucy mission's target, by analyzing light curves in Python

Summer 2022 **SIZLE: Smallsat to Image Zodiacal Light Above Ecliptic** *Cornell SmallSat Summer Design School*

- Served as Systems Engineer for a 20-person team developing a NASA SMEX mission concept to characterize the interplanetary dust cloud and diffuse zodiacal light
- Synthesized the Science Traceability Matrix (STM), leading flow-down of high-level scientific measurement goals into payload optical specifications, pointing requirements, and spacecraft bus trade studies
- Led interdisciplinary trade studies between science leads and engineering subsystems, validating that flight architectures fulfilled mission-critical baseline and threshold science requirements

2020 - 2021 **Low-thrust Trajectory Design to Outer-Planets**

- Implemented a c++ framework based on indirect method and genetic algorithm to design optimal low-thrust trajectories to outer planets using gravity assist

IIT Bombay Student Satellite Project - An interdisciplinary team of 40+ members across four subsystems

A student-run initiative to convert IIT Bombay to a Center of Excellence in Satellite and Space Technology

2018 - 2020 **Systems Engineer & Project Manager**

- Took over leadership during a critical time to ensure continuity post-launch of the first satellite along with the changes in the student satellite policies by the Indian Space Research Organisation (ISRO)
- Secured and maintained transparency of 1.5M INR funding every year in competitive grant applications
- Finalised the mission statement and payload from 30+ proposed payload ideas for Advitiy, a CubeSat concept to broadcast SSTV images to Indian villages, increasing space awareness
- Ensured implementation of Quality Assurance Practices; verified by ISRO scientists

2016 - 2017 **Mechanical Engineer** *Pratham, First Student Satellite of IIT Bombay, launched on 26th Sep.'16*

- Qualified satellite interface design through vibration testing at U. R. Rao Satellite Centre (URSC), Bengaluru, leveraging Solidworks and ANSYS Workbench to validate performance under PSLV launch specifications
- Developed testing plan of all onboard subsystems in vacuum undergoing temperatures of -60°C to 100°C for the environment testing of the flight model

Professional Service and Affiliations

2025 **Judge - Space Technology Payload Challenge** *NASA TechLeap 2025*

2024 - 2026 **Executive Secretary** *NASA ROSES Review Panels*

Oct 2024 **Co-chair at AAS DPS Meeting** *Lab Studies, Meteorites, and Instrument/Mission Development – Part 2*

2024 **Working Group Member** *Near-Earth Objects Workshops to Assess Reconnaissance for Planetary Defense*

Teaching & Mentoring Experience

Graduate Teaching Assistant *University of Maryland, College Park*

Spring 2025 Space Flight Dynamics - ENAE 404

Fall 2024 Spacecraft Navigation and Guidance - ENAE 441

- Designed and delivered active-learning recitations and formative assessments to improve conceptual understanding and problem-solving confidence
- Developed programming-based assignments in Python and MATLAB to connect dynamics and guidance theory with practical engineering workflows

2024 **Future Faculty Fellow** *A. James Clark School of Engineering, University of Maryland*

- Selected for UMD's Future Faculty Program, which prepares doctoral fellows for faculty careers through training in pedagogy, course design, active learning, assessment, and research mentoring

Graduate Teaching Assistant *IIT Bombay*

Spring 2020 Spaceflight Mechanics & Dynamics - AE240 & AE713

Fall 2019 Aerodynamics of Aerospace Vehicles - AE707

- Supported students in applying analytical foundations to aerospace design and problem-solving
- Facilitated problem-solving sessions and one-on-one mentoring for a diverse student group

2024 **Research Mentor** *Planetary Surfaces and Spacecraft Lab*

- Mentored first-year undergraduates and a high-school student on Brazil nut effect research in simulations of paramagnetic particles
- Provided structured mentoring with regular checkpoints, emphasizing inclusive team practices and clear technical communication

2021 **Team Mentor** *Terps in Space Mission 16, University of Maryland*

- Co-mentored four sophomores to design a space experiment selected for flight to the International Space Station

2018 - 2020 **Social Goal as Project Manager** *IIT Bombay Student Satellite Team*

- Structured & contributed to **Satellite 101** wiki as a comprehensive resource for new satellite teams, covering project management, systems engineering, and technical design
- Organised **Ground-station Workshop** for 50+ students from 15+ colleges of India; Taught *Systems Engineering Principles and Useful Team Building Practices*

Invited Talks and Public Outreach

2026 - present **Space Rocks - smallbodies.substack.com** *Newsletter on small bodies and planetary defense*

2021 - present **Indian Space Ecosystem Blog - blog.aerospacenerd** *with 650+ subscribers*

2026 **Planetary Regolith Dynamics and Exploration of Metallic Asteroids** *GaTech Space Week*

2026 **How Magnetic Interactions Alter the Fluidity of Metallic Asteroid Regolith** *Burgers Symposium, JHU*

2026 **Lightning Talk** *At 34th Meeting of NASA's Small Bodies Assessment Group (SBAG)*

2025 **Unlocking Metallic Worlds: Granular Mechanics on Asteroid Psyche** *Aerospace Dept, IIT Bombay*

2023 **Path to Graduate School** *Alumni Core Talks for Aerospace Undergraduates at IIT Bombay*

2020 **How to start a student satellite team?** *Somaiya Space Conclave*

2018 **Student Satellite Project Exhibition to the Prime Minister, India** *Silver Jubilee of the IIT Bombay*

2017 **Pratham: IITB Student Satellite** *Technology Forum on Student Micro/nano-satellites (Harbin, China)*

References

Prof. Christine Hartzell
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